# Critical Analysis

# On *Variational Tensor Neural Networks for Deep Learning*

by Saeed S. Jahromi and Román Orús — *Scientific Reports*, August 2024

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## 1 Introduction and State of the Art

The article analysed here, *Variational Tensor Neural Networks for Deep Learning*, was published in *Scientific Reports* in August 2024 by Saeed S. Jahromi and Román Orús, with co-authorship from Multiverse Computing, a quantum-inspired software company.

The authors frame their work as a response to two long-standing limitations of deep learning: the explosion of parameters in large networks, and the *opacity* of trained weights, which reveal little about what a model has actually learned. Their solution is a Tensor Neural Network (TNN) trained with a quantum-physics-inspired algorithm called DMRG (Density Matrix Renormalization Group, originally introduced by White [2]).

To assess the genuine novelty of the paper, it must be placed inside an already mature research line. Stoudenmire and Schwab [3] showed that supervised classification could be performed by a single Matrix Product State trained with a DMRG-like sweep — the first explicit bridge between tensor networks and machine learning. Earlier still, Novikov *et al.* [4] introduced "tensorizing neural networks" using tensor-train decompositions, compressing fully-connected layers by up to four orders of magnitude with negligible accuracy loss. Gao *et al.* [5] extended this idea by compressing entire deep networks with Matrix Product Operators (MPOs); Patel *et al.* [6] used MPO-based TNNs to solve partial differential equations; Wang *et al.* [7] deployed tensor networks for anomaly detection. On the physics side, the survey of Verstraete *et al.* [8] and the practical introduction by Orús [9] had long established that tensor networks and quantum many-body states obey common variational principles — making the cross-pollination with neural networks natural. Tensorization *per se* is therefore not new, and DMRG-style training had already been demonstrated.

What sets the present paper apart is twofold. First, the algorithm is generalised from the single-layer, task-specific schemes above to a *deep, multilayer* architecture that freely mixes tensor and dense layers within a single training loop. Second — and far more significant — the authors elevate the training procedure itself to a diagnostic tool: the DMRG sweep is reformulated so that the entanglement spectrum of the trained weights becomes directly readable at every step. The real contribution is therefore not scalability, as the abstract suggests, but *interpretability* — understood here as the ability to attach a humanly meaningful, quantitative explanation to a model's internal state, rather than merely observing that it maps inputs to outputs correctly. This is the angle on which the present analysis will focus.

## 2 Concept and Mathematical Formulation

A tensor network factorises a large weight matrix into a chain of small connected tensors. Originating in quantum physics for representing many-body wave functions, it doubles as a compression scheme

in deep learning: only the components carrying meaningful correlations are retained, and the rest discarded.

The authors use the **Matrix Product Operator** (MPO). A weight matrix $M \in \mathbb{R}^{p^2 \times p^2}$ is reshaped to rank 4 and decomposed via the Singular Value Decomposition (SVD):

$$M = USV^\dagger = LR, \qquad L = U\sqrt{S}, \qquad R = \sqrt{S}V^\dagger. \tag{1}$$

The diagonal matrix $S$ collects the singular values $\{\lambda_i\}$, and truncating to the $\chi_t \leq \chi$ largest of them defines the *bond dimension*. This parameter controls how much information passes between tensors — and, crucially, it quantifies the *entanglement* (correlation) between different parts of the network.

The full tensorisation procedure is summarised in Fig. 1. Panel (a) shows the classic dense layer $y_p = \sigma(Wx + b)$; panel (b) shows how the $216 \times 216$ weight matrix $W$ is reshaped and split, via successive SVDs, into three smaller MPO tensors $w_1, w_2, w_3$ linked by virtual bonds of dimension $\chi$; panel (c) shows the resulting TNN forward pass. The key visual insight is that the wide blue lines of width 216 are replaced by three narrow red legs of width 6 each, with controlled black bonds $\chi$ in between — compression and interpretability arise from the *same* structural change.

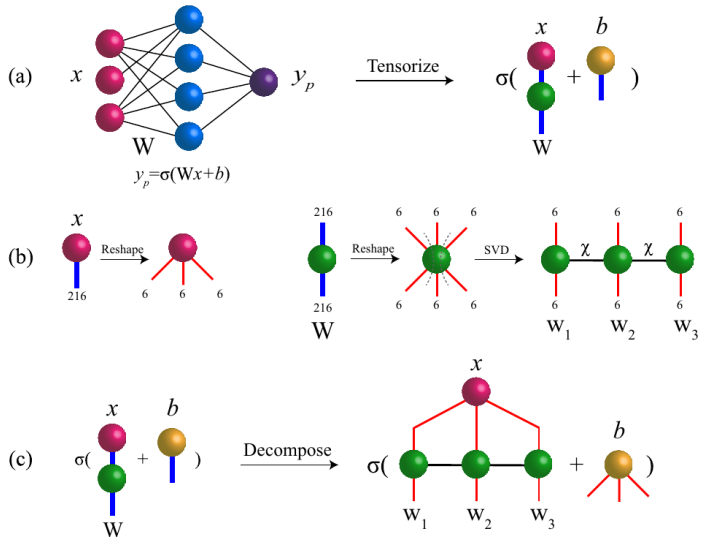



**Figure 1.** Tensorisation of a dense layer into a TNN (reproduced from Fig. 5 of [1]). (a) Standard NN → tensor diagram. (b) SVD-based decomposition of the weight matrix into three MPO tensors. (c) Resulting TNN forward pass.

The forward pass replaces the usual $y_p = \sigma(Wx + b)$ by the contracted form

$$y_p = \sigma(\mathrm{tTr}[x_{ijk\ldots}\, w_i w_j w_k \ldots] + b_{ijk\ldots}), \tag{2}$$

where tTr denotes the tensor trace over virtual indices.

For training, the authors adapt DMRG — a standard tool in computational quantum physics. Instead of optimizing all weights globally as Adam or SGD would, DMRG sweeps through tensor *pairs*, contracts each pair $w_i, w_{i+1}$ into a *bond tensor* $B_{i,i+1}$, and updates it locally:

$$B'_{i,i+1} = B_{i,i+1} - \alpha\, \Delta B_{i,i+1}, \qquad \Delta B = -\frac{\partial L_{\mathrm{MSE}}}{\partial B} = \frac{2}{N_s} \sum_{n=1}^{N_s} (y_j - y_p)\, M, \tag{3}$$

with $M$ the contraction of the network excluding $B$. The updated bond tensor is re-split by SVD,

$B' = U'S'V'^{\dagger}$, and truncated — which both controls the parameter count and *exposes the singular values at every step*. This is the key technical move: the two-site update is what makes entanglement directly observable. A one-site sweep, as in Novikov [4], would not produce this spectrum.

The internal correlations of the network are then summarised by the **entanglement entropy**

$$S = -\sum_i \lambda_i^2 \log \lambda_i^2, \tag{4}$$

computed at every virtual bond. A product-state-like model ($S \approx 0$) signals weak parameter correlation; large $S$ signals a strongly correlated, expressive representation. This is precisely where the interpretability gain becomes concrete. Where a conventional network exposes only a scalar loss — a measure of how well it fits, but mute on *what* it has internally organised — the entropy $S$ is a number read directly off the trained weights that characterises the correlation structure they encode. The opacity described in the introduction is thus not lifted by an external probing tool applied after training, but by a quantity intrinsic to the architecture and available throughout it. In short, two-site DMRG combined with MPO layers turns **correlation into a measurable quantity**. The framework further permits hybrid stacks of `TNLayer` and `Dense` layers, making the construction modular and practically usable.

## 3 Validation

The authors benchmark their method across four families of tasks — linear and non-linear regression, classification of Gaussian blobs and spirals, and MNIST image recognition. Table 1 collects the quantitative outcomes; the most diagnostic of them — and the cleanest evidence for the interpretability claim — come from the two-dimensional classifiers.

| **Task** | **Architecture** | **Accuracy / Fit** | **Entropy $S$** |
|---|---|---|---|
| Linear regression | 1 TNLayer ($N_{\mathrm{MPO}} = 6$) | $a_p = 0.57,\ b_p = 0.19$ [1] | — |
| Non-linear regression | 3 TNLayers (Linear/Sigmoid) | near-perfect $\sin x$ fit | — |
| Gaussian blobs | 1 TNLayer + 2 Dense | 99% | $\approx 0$ |
| Spiral distribution | 1 TNLayer + 2 Dense | 95% | $\approx 0.7$–$0.8$ |
| MNIST (TNN, 1 epoch) | 1 TNLayer ($N_{\mathrm{MPO}} = 12$) | 95.8% | — |
| MNIST (classical, 1 epoch) | Dense baseline (Adam) | 94.2% | — |
| MNIST (classical, 60 epochs) | Dense baseline (Adam) | 97.6% | — |

**Table 1.** Summary of the benchmark results reported in [1]. The entropy column is reported only for the classification tasks, where the entanglement structure of the trained weights was monitored. True line parameters: $a = 0.55$, $b = 0.20$.

The regression and blob tasks confirm that the TNN fits standard data accurately, but they are not where the contribution lies. The decisive comparison is between the two classifiers. On the **Gaussian blobs**, the model reaches 99% accuracy and the entanglement entropy collapses to nearly zero across all bond dimensions $\chi$, faithfully reflecting the clean separation of the two clusters. On the **spiral distribution**, the picture changes qualitatively. Fig. 2 shows the trained decision boundary winding cleanly through three interlocking arms, with an overall accuracy near 95%. What matters for the interpretability claim, however, is not this accuracy but the behaviour of the entanglement entropy reported by the authors: while the loss drops monotonically for all bond dimensions $\chi \in \{6, 10, 16\}$, the entropy $S$ does *not* decay to zero. Instead it converges to a non-trivial plateau around $S \approx 0.7$–$0.8$.

[1] Predicted slope and intercept, against true values $a = 0.55$, $b = 0.20$.

This non-vanishing entropy is the algebraic signature that the trained MPO weights are genuinely entangled — precisely the situation one expects when the input data are geometrically intertwined and cannot be separated by any product-state model. The internal entanglement measure thus *mirrors the geometric structure of the data* — well-separated data → low $S$; intertwined data → high $S$, as the entropy column of Table 1 makes explicit. Hence, reading a single scalar off the trained network already gives us actual insights about the data it has learned, which a standard dense network could not deliver.

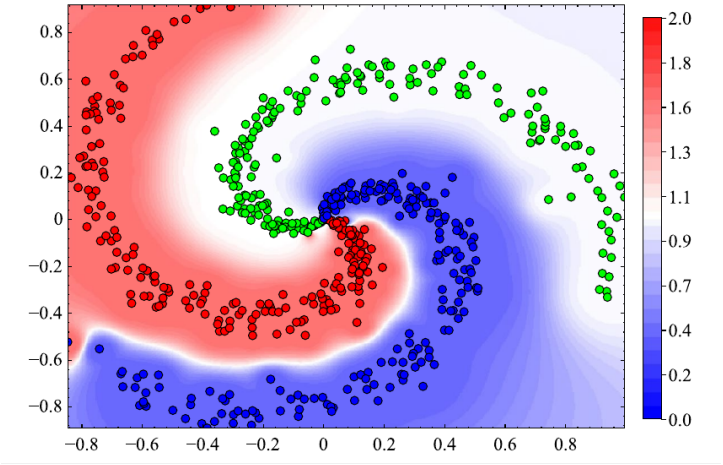



**Figure 2.** Ternary classification of a random spiral distribution. Filled circles are training data; red, white, blue shaded regions are the decision boundary obtained by the TNN. From Fig. 13 of [1].

On **MNIST**, the TNN reaches 95.8% test accuracy after a single training epoch at $N_{\mathrm{MPO}} = 12$, marginally above the 94.2% obtained by a classical fully-connected baseline trained for one epoch with Adam (Table 1). This single-epoch comparison is fragile: the same classical baseline climbs to 97.6% once trained to convergence over 60 epochs, and the earlier TNN approach of Stoudenmire & Schwab [3] performs comparably. More damningly, the authors themselves report that the DMRG-based training time per epoch grows *exponentially* with the number of MPO tensors $N_{\mathrm{MPO}}$, against roughly ten seconds for the Adam-trained baseline. The MNIST result is therefore best read as a proof of concept — which the authors openly acknowledge — rather than a claim of competitive performance.

## 4 Conclusion

Stripped of the abstract's framing, the genuine contribution of Jahromi and Orús is not improved scalability — prior MPO-based tensorization [4, 5] had already addressed that — but the coupling of *two-site* DMRG with multilayer MPO architectures, which makes the entanglement spectrum a first-class observable during training. Through the singular values exposed at each sweep and the entropy $S = -\sum_i \lambda_i^2 \log \lambda_i^2$, the network's internal correlation structure becomes a directly measurable quantity, aligned with the geometric complexity of the data, as Figures 2 and **??** make visible. This is the precise sense in which the work advances interpretability: it replaces the opaque, loss-only view of a trained network with an intrinsic, quantitative reading of what the model has organised internally. In an era dominated by large language models whose internal representations remain almost entirely opaque, such a reading offers interpretability research a rare quantitative anchor — rooted in the well-developed mathematical language of quantum information.

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